

Colorimetric Method for Dissolved Iron (Fe^{2+} / Fe^{tot}) Analysis

Claudia Mazur

Updated for SERC Biogeochem 12/19/22 by Alia Al-Haj

Adapted from Stookey, 1970, Violler *et al.*, 2000, Charbonnier & Anschutz 2021, Gong 2020

Principle: in neutral buffered medium ($4 < \text{pH} < 9$), the Fe^{2+} reacts with ferrozine to form a stable purple complex. The waters intended for the analysis of dissolved iron are acidified in order to maintain the ferrous ions in dissolved form. Some of these reduced ions can however be oxidized to Fe^{3+} , which is in dissolved form in an acidic medium. Ascorbic acid is therefore used during the assay to reduce ferric ions to ferrous ions (Viollier *et al.*, 2000). The percentage error of this method is approximately 5%.

Sample prep: Samples are filtered using $0.2\ \mu\text{m}$ polyethersulfone membranes and acidified with 1/100 v/v 1N HCl to bring the pH around 2. Samples are kept in the fridge at $\sim 4^\circ\text{C}$ in the dark.

Assay Protocol in brief:

1. Use glass volumetric flasks and sample tubes washed **in trace metal grade HCl**
2. Make Reagents (see below)
3. Make Standards
 - a. Standards can be prepared ahead of time, stored in the dark and in the fridge at $\sim 4^\circ\text{C}$.
4. Make Plate (see below)
 - a. First plate of the day must be standards only to double check curve
5. Run plate on microplate reader

Chemicals Required:

Chemical	CAS #	Supplier	SKU #	Precautions
Ammonium acetate	631-61-8	Fisher Scientific		Use in well ventilated hood
Ferrozine	50-81-7	Fisher Scientific	AC410570050	Use in well ventilated hood
Trace metal grade hydrochloric acid	7647-01-0	Fisher Scientific	A508-4	Use in well ventilated hood
Ammonium Hydroxide	1336-21-6	Fisher Scientific	A512-P500	Use in well ventilated hood
Ferrous Ammonium Sulfate Hexahydrate	7783-85-9	Fisher Scientific	LC144601	Use in well ventilated hood
Ferric Chloride Hexahydrate		Fisher Scientific		

Reagent Preparation:

(A) Ferrozine

1. Ammonium Acetate solution
 - a. Dissolve 38.54 g of ammonium acetate in 100 mL DI water
2. Ferrozine solution
 - a. Dissolve 0.514 g ferrozine in 100 mL DI water
3. Create a 50-50 mixture of the ammonium acetate (1) and ferrozine (2) solutions to make the final reagent. Store in the fridge in the dark for several months.

(B) Reducing Agent (1.4 M hydroxylamine hydrochloride in 2M HCl)

1. Add about 60 mL of DI water to a 100 mL volumetric flask.
2. Pipette 16.53 mL concentrated trace metal grade HCl into the 100 mL flask.
3. Add 9.729 g hydroxylamine hydrochloride to the flask.
4. Fill flask to the line with DI and mix well.

(C) Buffer (10 M ammonium acetate adjusted to pH 9.5)

1. Dissolve 77.08 g ammonium acetate in 100 mL DI.
2. Adjust pH to 9.5 with ammonium hydroxide (takes ~20 mL). (Use nutrient lab pH probe!! – Do not use a pH probe you use for field samples for chemicals)

Standard Preparation:

1. 0.5 M HCl
 - Add about 250 mL DI water to a 1000 mL volumetric flask.
 - Add 41.32 mL of trace metal grade HCl to the flask.
 - Fill the flask to the line.
2. 10 mM Fe(II) Stock
 - Add about 50 mL 0.5 M HCl to a 100 mL volumetric flask.
 - Add 0.3921 g Ferrous (Fe(II)) ammonium sulfate hexahydrate to the flask.
 - Fill the flask to the line with 0.5 M HCl.
3. 20 mM Fe(III) Stock
 - Add about 50 mL 0.5 M HCl to a 100 mL volumetric flask.
 - Add 0.5406 g Ferric (Fe(III)) chloride hexahydrate to the flask.
 - Fill the flask to the line with 0.5 M HCl.
4. Make the highest standard (Std A) in a volumetric flask and then perform a serial dilution from the highest standard into scintillation vials
5. Dilute standards

High Fe (II) Curve:

Standard	[Fe(II)] (uM)	0.5 M HCl (mL)	Standard (mL)	Total Volume
A	500	95	5 mL Fe(II) Stock	100 mL
B	250	10	10 mL Std A	20 mL
C	100	16	4 mL Std A	20 mL
D	75	17	3 mL Std A	20 mL
E	50	18	2 mL Std A	20 mL
F	25	19	1 mL Std A	20 mL
G	10	19.6	0.4 mL Std A	20 mL
H	2	19.92	0.08 mL Std A	20 mL
0	0	20.0	0	20 mL

High Fe (III) Curve:

Standard	[Fe(II)] (uM)	0.5 M HCl (mL)	Standard (mL)	Total Volume
A	500	97.5	2.5 mL Fe(III) Stock	100 mL
B	250	10	10 mL Std A	20 mL
C	100	16	4 mL Std A	20 mL
D	75	17	3 mL Std A	20 mL
E	50	18	2 mL Std A	20 mL
F	25	19	1 mL Std A	20 mL
G	10	19.6	0.4 mL Std A	20 mL
H	2	19.92	0.08 mL Std A	20 mL
0	0	20.0	0	20 mL

Low Fe (II) Curve:

Standard	[Fe(II)] (uM)	0.5 M HCl (mL)	Standard (mL)	Total Volume
A	60	99.4	0.6 mL Fe(II) Stock	100 mL
B	45	5	15 mL Std A	20 mL
C	30	10	10 mL Std A	20 mL
D	18	14	6 mL Std A	20 mL
E	9	17	3 mL Std A	20 mL
F	6	18	2 mL Std A	20 mL
G	3	19	1 mL Std A	20 mL
H	0.6	19.8	0.2 mL Std A	20 mL
0	0	20.0	0	20 mL

Low Fe (III) Curve:

Standard	[Fe(II)] (uM)	0.5 M HCl (mL)	Standard (mL)	Total Volume
A	60	99.4	0.3 mL Fe(III) Stock	100 mL
B	45	5	15 mL Std A	20 mL
C	30	10	10 mL Std A	20 mL
D	18	14	6 mL Std A	20 mL
E	9	17	3 mL Std A	20 mL
F	6	18	2 mL Std A	20 mL
G	3	19	1 mL Std A	20 mL
H	0.6	19.8	0.2 mL Std A	20 mL
0	0	20.0	0	20 mL

Notes:

Matrix Effect Standards: NO MATRIX EFFECT BECAUSE SODIUM BICARBONATE HAS VERY HIGH Fe

Spikes: Add 55 uL of Std A to well with sample.

Well Organization: Use the IronStd_Microplate_Template and Iron_Microplate_Template to organize samples in wells.

Plate Instructions:

1. Add 160 μ L of standard/sample to well
2. Add 16 uL **Ferrozine** to well
3. Immediately Measure the absorbance in each well at 562 nm with a microplate reader.
 - a. Read samples in SERC Terrestrial Ecology Lab
 - i. Turn on spec
 1. Door will open
 - ii. Log into computer
 - iii. Open Gen 5 software
 - iv. "Create a new item" $\rightarrow$ "experiment"
 - v. Select "default protocol", press "ok"
 - vi. From plate menu at the top (bar), select "read"
 1. Little plate icon
 - vii. Add step "shake" from the left
 1. Shake on medium speed for 10 seconds
 - viii. Add step "read" from the left
 - ix. Change wavelength to 562 nm
 - x. Press "OK" then "OK"
 - xi. Press "read"
 - xii. Load the plate – match up A1 on holder to plate
 - xiii. Click "OK", door will close
 - xiv. After plate is read, under "data" click "562: to see numbers"
 - xv. Click excel button on top of file (next to name)
 - xvi. Save excel file under your folder
 - xvii. Save the run file in the software as well
4. Add 33 μ L of **Reducing Agent**
5. Mix the plate for 10 seconds
6. Let the plate sit in the dark for 10 minutes
7. Add 11 uL **Buffer**
8. Immediately Measure the absorbance in each well at 562 nm with a microplate reader.
 - a. Read samples in SERC Terrestrial Ecology Lab
 - i. Turn on spec
 1. Door will open
 - ii. Log into computer
 - iii. Open Gen 5 software

- iv. "Create a new item" → "experiment"
- v. Select "default protocol", press "ok"
- vi. From plate menu at the top (bar), select "read"
 1. Little plate icon
- vii. Add step "shake" from the left
 1. Shake on medium speed for 10 seconds
- viii. Add step "read" from the left
- ix. Change wavelength to 562 nm
- x. Press "OK" then "OK"
- xi. Press "read"
- xii. Load the plate – match up A1 on holder to plate
- xiii. Click "OK", door will close
- xiv. After plate is read, under "data" click "562: to see numbers"
- xv. Click excel button on top of file (next to name)
- xvi. Save excel file under your folder
- xvii. Save the run file in the software as well

References:

Anschutz, Pierre, and Céline Charbonnier. "Sampling pore water at a centimeter resolution in sandy permeable sediments of lakes, streams, and coastal zones." *Limnology and Oceanography: Methods* 19.2 (2021): 96-114.

Stookey, Lawrence L. "Ferrozine---a new spectrophotometric reagent for iron." *Analytical chemistry* 42.7 (1970): 779-781.

Viollier, E., et al. "The ferrozine method revisited: Fe (II)/Fe (III) determination in natural waters." *Applied geochemistry* 15.6 (2000): 785-790.

Gong, J. "Dissolved Fe(II/III) colorimetric assay using a plate reader (96-well plate). *Protocols.io* (2020) <https://dx.doi.org/10.17504/protocols.io.bd5fi83n>.